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Method and apparatus for retrieving units from a storage system

Abstract

A load handling device is disclosed for retrieving, transporting, and delivering containers in a storage system having a grid structure formed of two substantially perpendicular sets of rails or tracks located above stacks of containers. An exemplary load handling device can include a wheel assembly, a container receiving space, and a plurality of vehicle modules. The wheel assembly can include a first set of wheels to guide movement of the load handling device in a first direction and a second set of wheels to guide movement of the load handling device in a second direction. Each of the vehicle modules can include a container lifting means arranged to lift a container into the container receiving space and each of the vehicle modules can be moveable to adjust the spacing of the containers in the load handling device.

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Background/Summary

METHOD AND APPARATUS FOR RETRIEVING UNITS FROM A STORAGE SYSTEM (1)

The present invention relates to apparatus for retrieving units from a storage system. In particular, but not exclusively, the invention relates to robotic devices for handling storage containers or bins in a store comprising a grid of stacked units. (2) This application is a continuation of U.S. patent application Ser. No. 17/733,464, now U.S. Pat. No. 11,738,945, filed on Apr. 29, 2022, which is a continuation of U.S. patent application Ser. No. 16/338,205, now U.S. Pat. No. 11,345,543, filed on Mar. 29, 2019, which is a continuation of International Application No. PCT/EP2017/075018 filed on Oct. 2, 2017, which claims priority from UK Patent Application No GB1616597.9 filed on 30 Sep. 2016. The content of each prior application is hereby incorporated by reference in its entirety.

(1) Some commercial and industrial activities require systems that enable the storage and retrieval of a large number of different products. One known system for the storage and retrieval of items in multiple product lines involves arranging storage bins or containers on rows of shelves arranged in aisles. Each bin or container holds a plurality of products of one product type. The aisles provide access between the rows of shelves, so that the required products can be retrieved by operatives or robots that circulate in the aisles. It will be appreciated, however, that the need to provide aisle space to access the products means that the storage density of such systems is relatively low. In other words, the amount of space actually used for the storage of products is relatively small compared to the amount of space required for the storage system as a whole.

(2) In an alternative approach, which offers a significant improvement in storage density, containers are stacked on top of one another and the stacks are arranged in rows. The containers are accessed from above, removing the need for aisles between the rows and allowing more containers to be stored in a given space.

(3) Methods of handling containers stacked in rows have been well known for decades. In some such systems, for example as described in U.S. Pat. No. 2,701,065, free-standing stacks of containers are arranged in rows in order to reduce the storage volume associated with storing such containers while still providing access to a specific container if required. Access to a given container is made possible by providing relatively complicated hoisting mechanisms which can be used to stack containers and to remove given containers from stacks. The cost of such systems are, however, impractical in many situations and they have mainly been commercialized for the storage and handling of large shipping containers.

(4) The concept of using freestanding stacks of containers and providing a mechanism to retrieve and store specific containers has been developed further, for example as described in EP 0 767 113 B (Cimcorp). Cimcorp discloses a mechanism for removing a plurality of stacked containers using a robotic load handler in the form of a rectangular tube which is lowered around the stack of containers, and which is configured to be able to grip a container at any level in the stack. In this way, several containers can be lifted at once from a stack. The movable tube can be used to move several containers from the top of one stack to the top of another stack, or to move containers from a stack to an external location and vice versa. Such systems can be particularly useful where all of

the containers in a single stack contain the same product (known as a single-product stack). The load handler can be used to move containers between single-product stacks, for example to add a plurality of containers containing a single3(a) type of product to the store, and to pick up one or more containers from two or more single-product stacks to create a multi-product output stack. An example of this is the picking of vegetable crates in a central warehouse to create a multi-product order for delivery to retail stores.

(5) In the system described in Cimcorp, the height of the tube has to be as least as high as the height of the largest stack of containers, so that that the highest stack of containers can be extracted in a single operation. Accordingly, when used in an enclosed space such as a warehouse, the maximum height of the stacks is restricted by the need to accommodate the tube of the load handler. Furthermore, the system is not well adapted for the selection of a single container from a multi-product stack.

(6) Online retail businesses selling multiple product lines, such as online grocers and supermarkets, require systems that are able to store tens or even hundreds of thousands of different product lines. The use of single-product stacks in such cases can be impractical, since a very large floor area would be required to accommodate all of the stacks required. Furthermore, it can be desirable only to store small quantities of some items, such as perishables or infrequently-ordered goods, making single-product stacks an inefficient solution.

(7) Accordingly, for some applications, the use of multi-product stacks, in which the containers making up each stack may hold different products, is favoured in order to maximise the storage density of the system. The stored items must remain accessible reasonably quickly and easily, so that a plurality of different items required to fulfil a customer order can be picked from the storage system in an efficient way, even if some of the items required are stored in a lower level of a stack, underneath several other containers.

(8) International patent application WO 98/049075A (Autostore), the contents of which are incorporated herein by reference, describes a system in which multi-product stacks of containers are arranged within a frame structure. A system of this type is illustrated schematically in FIGS. **1** to **4** of the accompanying drawings.

(9) As shown in FIGS. **1** and **2**, stackable containers, known as bins **10**, are stacked on top of one another to form stacks **12**. The stacks **12** are arranged in a grid frame structure **14** in a warehousing or manufacturing environment. FIG. **1** is a schematic perspective view of the frame structure **14**, and FIG. **2** is a top-down view showing a stack **12** of bins **10** arranged within the frame structure **14**. Each bin **10** typically holds a plurality of product items (not shown), and the product items within a bin **10** may be identical, or may be of different product types depending on the application.

(10) The frame structure **14** comprises a plurality of upright members **16** that support horizontal members **18**, **20**. A first set of parallel horizontal members **18** is arranged perpendicularly to a second set of parallel horizontal members **20** to form a plurality of horizontal grid structures supported by the upright members **16**. The members **16**, **18**, **20** are typically manufactured from metal. The bins **10** are stacked between the members **16**, **18**, **20** of the frame structure **14**, so that the frame structure **14** guards against horizontal movement of the stacks **12** of bins **10**, and guides vertical movement of the bins **10**.

(11) The top level of the frame structure **14** includes rails **22** arranged in a grid pattern across the top of the stacks **12**. Referring additionally to FIGS. **3** and **4**, the rails **22** support a plurality of robotic load handling devices **30**. A first set **22a** of parallel rails **22** guide movement of the load handling devices **30** in a first direction (X) across the top of the frame structure **14**, and a second set **22b** of parallel rails **22**, arranged perpendicular to the first set **22a**, guide movement of the load handling devices **30** in a second direction (Y), perpendicular to the first direction. In this way, the rails **22** allow movement of the load handling devices **30** laterally in two dimensions in the horizontal X-Y plane, so that a load handling device **30** can be moved into position above any of the stacks **12**.

(12) The load handling devices **30** are further described in Norwegian patent number **317366**, the contents of which are incorporated herein by reference. FIGS. **3a** and **3b** are schematic perspective views of a load handling device **30** from the rear and front, respectively, and FIG. **3c** is a schematic front perspective view of a load handling device **30** lifting a bin **10**.

(13) Each load handling device **30** comprises a vehicle **32** which is arranged to travel in the X and Y directions on the rails **22** of the frame structure **14**, above the stacks **12**. A first set of wheels **34**, consisting of a pair of wheels **34** on the front of the vehicle **32** and a pair of wheels **34** on the back of the vehicle **32**, is arranged to engage with two adjacent rails of the first set **22a** of rails **22**. Similarly, a second set of wheels **36**, consisting of a pair of wheels **36** on each side of the vehicle **32**, is arranged to engage with two adjacent rails of the second set **22b** of rails **22**. Each set of wheels **34**, **36** can be lifted and lowered, so that either the first set of wheels **34** or the second set of wheels **36** is engaged with the respective set of rails **22a**, **22b** at any one time.

(14) When the first set of wheels **34** is engaged with the first set of rails **22a** and the second set of wheels **36** is lifted clear from the rails **22**, the wheels **34** can be driven, by way of a drive mechanism (not shown) housed in the vehicle **32**, to move the load handling device **30** in the X direction. To move the load handling device **30** in the Y direction, the first set of wheels **34** is lifted clear of the rails **22**, and the second set of wheels **36** is lowered into engagement with the second set of rails **22a**. The drive mechanism can then be used to drive the second set of wheels **36** to achieve movement in the Y direction.

(15) The load handling device **30** is equipped with a crane device **40**. The crane device **40** comprises a cantilever arm **42** that extends laterally from the top of the vehicle **32**. A gripper plate **44** is suspended from the cantilever arm **42** by four cables **46**. The cables **46** are connected to a winding mechanism (not shown) housed within the vehicle **32**. The cables **46** can be spooled in or out from the cantilever arm **42**, so that the position of the gripper plate **44** with respect to the vehicle **32** can be adjusted in the Z direction.

(16) The gripper plate **44** is adapted to engage with the top of a bin **10**. For example, the gripper plate **44** may include pins (not shown) that mate with corresponding holes (not shown) in the rim that forms the top surface of the bin **10**, and sliding clips (not shown) that are engageable with the rim to grip the bin **10**. The clips are driven to engage with the bin **10** by a suitable drive mechanism housed within the gripper plate **44**, which is powered and controlled by signals carried through the cables **46** themselves or through a separate control cable (not shown).

(17) To remove a bin **10** from the top of a stack **12**, the load handling device **30** is moved as necessary in the X and Y directions so that the gripper plate **44** is positioned above the stack **12**. The gripper plate **44** is then lowered vertically in the Z direction to engage with the bin **10** on the top of the stack **12**, as shown in FIG. **3c**. The gripper plate **44** grips the bin **10**, and is then pulled upwards on the cables **46**, with the bin **10** attached. At the top of its vertical travel, the bin **10** is accommodated beneath the cantilever arm **42** and is held above the level of the rails **22**. In this way, the load handling device **30** can be moved to a different position in the X-Y plane, carrying the bin **10** along with it, to transport the bin **10** to another location. The cables **46** are long enough to allow the load handling device **30** to retrieve and place bins from any level of a stack **12**, including the floor level. The vehicle **32** is sufficiently heavy to counterbalance the weight of the bin **10** and to remain stable during the lifting process. The weight of the vehicle **32** may be comprised in part of batteries that are used to power the drive mechanism for the wheels **34**, **36**.

(18) As shown in FIG. **4**, a plurality of identical load handling devices **30** are provided, so that each load handling device **30** can operate simultaneously to increase the throughput of the system. The system illustrated in FIG. **4** includes two specific locations, known as ports **24**, at which bins **10** can be transferred into or out of the system. An additional conveyor system (not shown) is associated with each port **24**, so that bins **10** transported to a port **24** by a load handling device **30** can be transferred to another location by the conveyor system, for example to a picking station (not shown). Similarly, bins **10** can be moved by the conveyor system to a port **24** from an external

location, for example to a bin-filling station (not shown), and transported to a stack **12** by the load handling devices **30** to replenish the stock in the system.

(19) Each load handling device **30** can lift and move one bin **10** at a time. If it is necessary to retrieve a bin **10** ("target bin") that is not located on the top of a stack **12**, then the overlying bins **10** ("non-target bins") must first be moved to allow access to the target bin **10**.

(20) Each of the load handling devices **30** is under the control of a central computer. Each individual bin **10** in the system is tracked, so that the appropriate bins **10** can be retrieved, transported and replaced as necessary.

(21) The system described with reference to FIGS. **1** to **4** has many advantages and is suitable for a wide range of storage and retrieval operations. In particular, it allows very dense storage of product, and it provides a very economical way of storing a huge range of different items in the bins **10**, while allowing reasonably economical access to all of the bins **10** when required for picking.

(22) For high-volume systems in which speed of operation is critical, it is important to maximise the performance of each of the load handling devices, in terms of speed of operation, battery life, reliability, lifting capacity, stability and so on. It may therefore be desirable to provide load-handling devices that offer improved performance in one or more of these areas.

(23) It may also be desirable to increase the number of load handling devices in use at any one time, to allow an increase in the speed with which items can be retrieved from the storage system. For example, the Applicant's co-pending International Patent Application No.

PCT/GB2013/051215, the content of which is incorporated herein by reference, describes a storage system in which a plurality of each of two different types of load handling device are provided.

One type of load handling device is adapted to lift a plurality of bins from a stack in one operation, to allow a target bin in the stack to be accessed by a single-bin load handling device of the second type. In such cases, it may be desirable to reduce the size of the load handling devices in order to minimise instances in which the optimum movement path for one device is hindered by the presence of other devices.

(24) Load handling devices of the type described above are expensive and it is an object of the present invention to reduce the electronics and control mechanisms by providing a single load handling device capable of lifting and carrying multiple containers.

(25) It is against this background that the present invention has been devised.

(26) According to the invention there is provided a load handling device for lifting and moving containers stacked in a storage system comprising a plurality of rails or tracks arranged in a grid pattern above the stacks of containers, the grid pattern comprising a plurality of grid spaces and each stack being located within a footprint of only a single grid space, the load handling device being configured to move laterally on the rails or tracks above the stacks, the load handling device comprising a container receiving space located above the rails or tracks in use and a plurality of vehicle modules, the modules comprising container lifting means arranged to lift a corresponding number of containers as vehicle modules in to the container receiving space.

(27) According to the invention there is further provided a storage system comprising: a first set of parallel rails or tracks and a second set of parallel rails or tracks extending substantially perpendicularly to the first set of rails or tracks to form a grid comprising a plurality of grid spaces; a plurality of stacks of containers located beneath the rails or tracks and arranged such that each stack occupies a footprint of a single grid space; a load handling device as arranged to move laterally above the stacks on the rails, the load handling device comprising a container receiving recess located above the rails and a plurality of vehicle modules, the vehicle modules comprising lifting means the lifting means arranged to lift a plurality of containers from a plurality of stacks into the container receiving space.

(28) According to the invention there is further provided a method of transferring a plurality of containers located within a plurality of adjacent stacks in a storage system from the storage system

to a shipping trolley, the method comprising the steps of: retrieving a plurality of containers from a correspondingly numbered plurality of stacks; transporting the containers in a load handling device, the load handling device comprising a plurality of vehicle modules; delivering the containers to a port location; lowering the containers on to a container receiving platform; adjusting the spacing of the containers using adjustment means; and loading the containers on to the shipping trolley.

(29) A load handling device according to an embodiment of the invention includes a container-receiving space into which a plurality of containers can be lifted. The container-receiving space is arranged beneath a corresponding number of vehicle modules as containers to be lifted, and components such as power components, control components, drive components and lifting components are housed.

(30) By arranging the bulky components of the load handling device above the container-receiving space, the footprint of the load handling device is reduced compared to the cantilever designs shown in FIGS. 3a to 3c and described in NO317366, in which the bulky components are housed in a vehicle module disposed to one side of the container-receiving space. Advantageously, the load handling device of the present invention occupies the space above a corresponding number of stacks in the frame as vehicle modules and containers to be lifted.

(31) This means that, by virtue of the invention, the efficiency of operation of the storage system can be improved, because the footprint of the load handling device allows more containers to be lifted in a single operation than a load handling device capable of lifting only one container.

(32) The load handling device preferably includes a set of wheels for supporting the load handling device above the stacks. For example, lateral movement of the load handling device may be guided by rails disposed above the frame. The rails may be arranged in a grid pattern, allowing two-dimensional movement of the load handling device in the horizontal plane. The wheels may engage with the rails. Two sets of wheels may be provided, with one set being arranged to engage with a first set of rails to guide movement of the second handling device in a first direction, and another set being arranged to engage with a second set of rails to guide movement of the second handling device in a second direction.

(33) In an embodiment of the invention, the wheels are arranged at the periphery of the container-receiving space. The wheels may be driven by one or more motors housed in the vehicle module. Drive may be transferred from the motors in the vehicle module to the wheels by drive transfer means disposed around the container-receiving space. For example, the drive transfer means may comprise a suitable arrangement of pulleys and drive belts.

(34) One or both set of wheels may be configured to be raised and lowered with respect to the other set of wheels. One or more wheel lift motors or other wheel lift devices may be housed in the vehicle module for this purpose.

(35) The vehicle module may house a plurality of winches or crane devices for lifting a corresponding number of containers as vehicle modules into the container-receiving space. The crane devices may include one or more motors for lifting the containers, and the or each motor of the or each crane device may be housed in the or each vehicle module.

(36) Each crane device may include a gripper device configured to grip a container from above. The gripper device may be suspended from cables that can be extended and retracted from the vehicle to move the gripper device vertically.

(37) In another embodiment, the load handling device is equipped with a lifting device arranged to lift a plurality of containers from the stack into the container-receiving space. The lifting devices may comprise a pair of lifting arms, in which case the or each lifting device may comprise a gripper device mounted between the ends of the arms and arranged to grip a container from above.

(38) In another embodiment, the or each lifting device comprises rods or cables arranged to engage with vertical channels formed in the side walls of the containers. The channels may be accessed by apertures in a top face of each container. In such an arrangement, vertically-extending spaces in the storage system are not necessary.

(39) The rods or cables may carry an anchor mechanism arranged to engage releasably with a container. For example, the anchor mechanism may comprise one or more laterally-extendable arms for engaging a surface of the container. The anchor mechanism may be operated remotely, for example by a wire that extends through a tubular bore of the rod or cable.

(40) A load handling device according to another embodiment of the invention comprises an upper part, a lower part including a container-receiving space, and a plurality of winch means for lifting containers into the container-receiving space. The winch means comprise winch motors which are housed in the upper part, above the container-receiving space. The lower part includes a wheel assembly to facilitate lateral movement of the load handling device with respect to the frame, and the upper part also includes at least one motor for driving one or more wheels of the wheel assembly.

(41) The lower part may comprise a frame structure for supporting the wheels of the wheel assembly. The frame structure may be arranged around the container-receiving space. For example, the container-receiving space may be bounded on four sides by the frame structure. One or more elements of the frame structure may be moveable to raise and lower a first set of the wheels with respect to a second set of the wheels, thereby to facilitate engagement of either the first set of wheels or the second set of wheels with a first or a second set of rails or tracks, respectively. The moveable elements of the frame structure may be driven by a motor housed in the upper part of the load handling device.

(42) The load-handling device of the invention is preferably a self-propelled robot vehicle.

(43) From another aspect, the invention resides in a storage system comprising a frame containing a plurality of stacks of containers, and one or more load handling devices as described above. Each load handling device occupies a plurality of grid spaces, corresponding to the area occupied by a plurality of stacks of containers.

(44) In another aspect, the invention comprises a storage system comprising a frame containing a plurality of stacks of containers, a first handling device capable of lifting a plurality of containers from a plurality of stacks in a single operation, and a second handling device capable of lifting a single container and moving the container laterally. The first and second handling devices are disposed above the frame and are independently moveable to access different stacks. The second handling device is of the type described above, but occupies a space corresponding to only one stack of containers.

(45) In this aspect, the provision of a first handling device capable of lifting a plurality of containers from a plurality of stacks in a single operation along with a second handling device capable of lifting a single container and moving the container laterally provides an optimum solution when seeking to retrieve and move a large number of containers. In such a case, only a single lifting operations need be carried out to retrieve the plurality of target containers, which greatly increases the speed and efficiency of the retrieval process compared to prior art arrangements in which only one container can be lifted at a time.

(46) The storage system may further comprise one or more port locations at which containers can be removed from and/or added to the storage system. The load handling device of the invention may be capable of transporting a target containers from a plurality of stacks to a port location. The containers may comprise open-topped bins. The containers may be arranged to interlock or engage with one another in the vertical direction when formed in a stack.

(47) In a typical application, multiple handling devices may be employed so that a large number of containers can be lifted and moved simultaneously in and around the system. The handling devices may be of different types, and may be selected to balance the cost and energy consumption of the system with the speed and flexibility of operation. One benefit of the present invention is that, because the load handling devices occupy the space above a plurality of stacks, the efficiency of a multiple-device system can be improved compared to prior art load handling device designs which occupy two or more stack spaces to lift a single container. The gain in efficiency may arise from

being able to accommodate more load handling devices in a given system, from optimising the routing of the device, or from a combination of these factors.

(48) Preferably the load handling device of the invention occupies a 2×2 grid space and is capable of retrieving four containers in a single operation. Preferably the load handling device comprises four vehicle modules arranged in a 2×2 arrangement above the container-retrieving space of the load handling device. Preferably each vehicle module comprises a winch or crane lifting device for engaging with a container at the top of a stack above which the load handling device is positioned.

(49) Preferably the load handling device carrying the plurality of containers in the container receiving space, may be transported to a position above a port location under the control of a communications and control system. Once above the port location, the winch or crane lifting means may lower the plurality of containers on to a container receiving platform, the platform comprising means for moving the lowered containers on to a container shipping trolley comprising a series of shelves in substantially a single movement. The process repeats such that a container shipping trolley may be filled a shelf at a time in order to be ready to be loaded on to a vehicle for onward shipment.

(50) Preferred and/or optional features of each aspect of the invention may be used, alone or in appropriate combination in the other aspects of the invention also.

Description

(1) FIG. 1 is a schematic perspective view of a frame structure for housing a plurality of stacks of bins in a known storage system;

(2) FIG. 2 is a schematic plan view of part of the frame structure of FIG. 1;

(3) FIGS. 3*a* and 3*b* are schematic perspective views, from the rear and front respectively, of a known load handling device for use with the frame structure of FIGS. 1 and 2, and FIG. 3*c* is a schematic perspective view of the known load handling device in use lifting a bin; and

(4) FIG. 4 is a schematic perspective view of a known storage system comprising a plurality of load handling devices of the type shown in FIGS. 3*a*, 3*b* and 3*c*, installed on the frame structure of FIGS. 1 and 2.

(5) Embodiments of the present invention will now be described, by way of example only, with reference to the remainder of the accompanying drawings, in which like reference numerals are used for like features, and in which:

(6) FIG. 5 is a schematic perspective view of a load handling device according to an embodiment of the invention;

(7) FIGS. 6 and 7 are schematic perspective views of the load handling device of FIG. 5, showing the arrangement of four vehicle modules within the load handling device;

(8) FIG. 8 is a schematic perspective view of a storage system comprising a load handling device in accordance with FIG. 5 installed on the frame structure of FIGS. 1 and 2, the load handling device being in situ above four stacks of containers of the type to be retrieved;

(9) FIGS. 9 and 10 are schematic perspective views of the storage system and load handling device of FIGS. 6 and 7 showing container lifting means associated with each vehicle module engaging with a top container in each stack;

(10) FIGS. 11 and 12 are side views of the load handling device and stacks of FIGS. 9 and 10, the lifting means engaged with containers;

(11) FIG. 13 is a schematic perspective view of one form of port location in a storage system showing a receiving platform and transfer mechanism for transferring delivered containers on to a shipping trolley;

(12) FIG. 14 is a schematic perspective view of the port location of FIG. 13 showing two containers being lowered on to the container receiving platform by the load handling device;

(13) FIGS. **15** and **16** are schematic perspective views of the vehicle modules within the load handling device, showing movement of two of the vehicle modules to enable the containers to be delivered on to the container receiving platform in a position enabling the container transfer mechanism to relocate the delivered containers on to the shipping trolley;

(14) FIGS. **17** and **18** are schematic perspective views of the load handling device in position above a port location in accordance with one form of the invention, showing the container lifting means disengaging from the containers delivered on to the container receiving platform;

(15) FIG. **19** is a schematic perspective view of the load handling device having completely disengaged from the delivered containers, the containers being located on the delivery platform;

(16) FIGS. **20** and **21** are schematic perspective views of the delivered containers in situ on the container receiving platform, delivered by the load handling device of one form of the invention, the delivered containers being transferred via transferring means on to the shipping trolley; and

(17) FIGS. **22** to **26** are schematic perspective views showing the load handling device, the vehicle modules and the shipping trolley loading system in accordance with a further form of the invention in which the vehicle module spacing within the load handling device remains constant and the container receiving platform comprises two portions, a first portion being moveable relative to the second portion such that in use, the vehicle modules lower the containers on to the platforms, a portion of the platform moving so as to enable the delivered containers to be transferred to the shipping trolley.

(18) FIG. **5** shows a load handling device **100** according to an embodiment of the invention. The load handling device **100** comprises a vehicle **102** equipped with a plurality of vehicle modules **103**, each module **103** comprising winch or crane mechanisms **104** to lift a storage container or bin **106**, also known as a tote, from above. The crane mechanisms **104** includes winch cables **108** and grabber plates **110**. The grabber plates **110** are configured to grip the top of the containers **106** to lift them from a stack **12** of containers **106** in a storage system of the type shown in FIGS. **1** and **2**.

(19) Referring also to FIGS. **6** and **7**, the vehicle **102** comprises an upper part **112** and a lower part **114**.

(20) The lower part **114** is fitted with two sets of wheels **116**, **118**, which run on rails provided at the top of the frame of the storage system. At least one wheel of each set **116**, **118** is driven to enable movement of the vehicle **102** in X- and Y-directions respectively along the rails. As will be explained below, one or both sets of wheels **116**, **118** can be moved vertically to lift each set of wheels clear of the respective rails, thereby allowing the vehicle **102** to move in the desired direction.

(21) The wheels **116**, **118** are arranged around the periphery of a cavity or recess **120**, known as a container-receiving recess, in the lower part **114**. The recess **120** is sized to accommodate the or each bin **106** when it is lifted by the crane mechanism **104**, as shown in FIGS. **11** and **12**. When in the recess **120**, the bin **106** is lifted clear of the rails beneath, so that the vehicle **102** can move laterally to a different location. On reaching the target location, for example another stack **12**, an access point in the storage system or a conveyor belt, the bins **106** can be lowered from the recess **120** and released from the grabber plates **110**.

(22) The upper part **112** of the vehicle **102** houses all of the significant bulky components of the load handling device, as shown in FIGS. **6** and **7**. The upper part **112** houses the battery and associated electronics, controllers and communications devices, motors for driving the wheels **116**, **118**, motors for driving the crane mechanism **104**, and other sensors and systems.

(23) In this way, the footprint of the vehicle **102** is larger than the size of the bins **106** contained within the recess **120** only enough to accommodate the wheels **116**, **118** either side of the recess **120**. In other words, the vehicle **102** occupies a plurality of grid spaces in the storage system equivalent only to the number of vehicle modules contained within the load handling device **100**.

(24) The load handling devices **100** of the invention can also offer improved stability, increased load handling capacity and reduced weight compared to the cantilever-type prior art load handling

devices **30**, because in the invention the load of the containers is suspended between the pairs of wheels on each side of the vehicle. In contrast, the prior-art devices **30** must have a relatively heavy vehicle module to counterbalance the load in the cantilever configuration.

(25) FIGS. **5** to **21** show one embodiment of the invention. The upper part **112** of the vehicle **102** houses four vehicle modules **103**, each module **103** having three main motors: a Z-drive motor used to raise and lower the winch cables **108**, which are wound onto spools mounted on drive shafts situated at opposite sides of the vehicle **102**; an X-drive motor which drives the first set of wheels **116**, and a Y-drive motor which drives the second set of wheels **118**. The upper part **112** of the vehicle also houses a battery to power the motors, and controllers, sensors and other components as described above with reference to FIGS. **6** and **7**.

(26) Drive is transferred from the X- and Y-drive motors to the respective sets of wheels **116**, **118** by means of belt drive mechanisms. The X-drive motor or motors drive pulleys connected to a short drive shaft that extends across the vehicle body. Drive is transferred from the short drive shaft to each wheel in the first set of wheels **116** by an X drive belt. The Y-drive motor drives a pulley connected to a long drive shaft that extends across the vehicle body in a direction perpendicular to the short drive shaft. Drive is transferred from the long drive shaft to each wheel in the second set of wheels **118** by a Y drive belt.

(27) The belt-driven wheels **116**, **118** are mounted at the bottom of the lower part **114** of the vehicle **102**. The use of drive belts to transfer drive from the motors to the wheels enables the motors to be mounted in the upper part **112** of the vehicle.

(28) In this embodiment, as an example only, the first set of wheels **116** can be raised clear of the rails or lowered onto the rails by means of a wheel positioning mechanism. One form of mechanism may operate as follows: the wheel **116** is mounted on an arm that is pivotally mounted at its outer end. An inner end of each arm is connected to the lower end of a respective linkage. The upper ends of both linkages are connected to the lower end of a common linkage. In turn, the upper end of the common linkage is connected to a lever arm that is moved by a motor. By operating the motor to draw the common linkage upwards, the first set of wheels **116** can be raised so that the second set of wheels **118** alone is engaged with the rails, allowing movement of the vehicle **102** in the Y-direction. By operating the motor to push the common linkage downwards, the first set of wheels **116** move downwards to engage with the rails and to lift the vehicle so that the second set of wheels **118** is lifted clear of the rails. The vehicle **102** can then move in the X-direction.

(29) The wheels **118** of the second set may be mounted to fixed T-pieces disposed at either end of the lower part **114** of the vehicle **102**.

(30) It will be appreciated that the wheels need not be arranged around the periphery of the cavity or recess. Indeed, multiple sets of wheels may be provided so as to take advantage of the tracks or rails defining the footprint of the stacks of containers or bins **10**.

(31) FIGS. **8**, **9** and **12** show the load handling device **100** with bins **106** being lifted into the recess **120**. FIG. **11** shows the load handling device **100** with the bin **106** beneath the device **100** and one grabber plate **110** about to engage with one target bin **106**. The wheels **116**, **118** and the associated support pieces, linkages and drive belts are arranged around the edges of the recess **120**, so that the upper part **112** of the vehicle **102** is solidly supported.

(32) Wheels suitable for use as one of the wheels **116**, **118** of the load handling device **100** may comprise a toothed central channel that forms a pulley for cooperating with a drive belt. The channel is bounded by two rubber tyres, which bear upon the rails in use. The wheel can be mounted to an arm by way of an axle (not shown) that extends through an axial hole in the wheel. This wheel design is compact and balanced, to minimise wear, and the tyres serve to keep the drive belt in alignment in use.

(33) FIG. **13** is a schematic perspective view of one form of port location in a storage system showing a container receiving platform **198** and transfer mechanism **200** for transferring delivered containers on to a shipping trolley **202**.

(34) FIG. 14 is a schematic perspective view of the port location of FIG. 13 showing two containers **106** being lowered on to the container receiving platform **198** by the vehicle modules **103** of the load handling device **100**. As can be seen in FIG. 14, the spacing of the containers **106** in the container receiving recess **120** does not match the spacing of the container receiving shelves of the container shipping trolley **202**.

(35) FIGS. 15 and 16 are schematic perspective views of the vehicle modules **103** within the load handling device **100**, showing movement of two of the vehicle modules **103** to enable the containers **106** to be delivered on to the container receiving platform **198** in a position enabling the container transfer mechanism **200** to relocate the delivered containers **106** on to the shipping trolley **202**.

(36) FIGS. 17 and 18 are schematic perspective views of the load handling device **100** in position above a port location in accordance with one form of the invention, showing the container lifting means **110** disengaging from the containers **106** delivered on to the container receiving platform **198**.

(37) FIG. 19 is a schematic perspective view of the load handling device **100** having completely disengaged from the delivered containers **106**, the containers **106** being located on the delivery platform **198**.

(38) FIGS. 20 and 21 are schematic perspective views of the delivered containers **106** in situ on the container receiving platform **198**, delivered by the load handling device **100** of one form of the invention, the delivered containers **106** being transferred via transferring means on to the shipping trolley **202**.

(39) In use, two of the vehicle modules **103** lower a first two containers **106** on to one portion of the container receiving platform **198**. Once the first two containers **106** are lowered on to the platform, the opposing two vehicle modules **103** move within the body of the load handling device to a position above a second portion on the container receiving platform **198**. The vehicle modules **103** lower the two containers **106** on the container lifting means on to the second portion of the platform **198**. Once all four of the containers **106** are in the correct position on the container receiving platform **198**, the grippers **110** are winched back in to the load handling device **100**, the vehicle modules reposition and the load handling device **100** is free to be moved under the control of the communications and control system to a different location.

(40) The four containers **106** on the container platform **198** are transferred by suitable transferring means **200** on to the shipping trolley **202**. The transferring means may comprise a push plate mechanism, a conveyor mechanism or any other suitable form of transferring mechanism for moving the containers **106** on to the shipping trolley **202**.

(41) It will be appreciated that in the embodiments of the invention described above, the load handling device is adapted to retrieve four containers **106** in a 2×2 arrangement but that any other arrangement of containers **106** and vehicle modules **103** may be used, for example 1×2, 1×3, 2×3, and 3×3 or more may be envisaged. In the case where a different number of containers **106** are envisaged, the container receiving platform would also need to be adapted to accept the corresponding number of containers **106**.

(42) It will be appreciated that in these different container-vehicle module arrangements, it may be especially advantageous for the wheels not to be arranged around the periphery of the cavity or recess. Indeed, it may be necessary for multiple sets of wheels to be provided so as to take advantage of the tracks or rails defining the footprint of the stacks of containers or bins **10**.

(43) FIGS. 22 to 26 are schematic perspective views showing the load handling device **100**, the vehicle modules **103** and the shipping trolley **202** loading system **200** in accordance with a further form of the invention in which the vehicle module **103** spacing within the load handling device **100** remains constant and the container receiving platform **198** comprises two portions. A first portion of the platform **198** is moveable relative to the second portion of the platform **198** such that in use, the fixed position vehicle modules **103** lower the containers **106** on to the portions of the platform

198 by means of the container lifting device and gripper **110** mechanism. A portion of the platform **198** moves so as to enable the delivered containers **106** to be transferred to the shipping trolley **202**. (44) It will be appreciated that the load handling device **100** and the container receiving platform **198** are acting so as to adjust the pitch or spacing of the containers **106** to enable mismatches between container stack **12** pitch and shipping trolley container pitch **110**. However, it will be appreciated similar method and apparatus may be used to compensate for a mismatch in spacing or pitch at other points in the process or system.

(45) It will be appreciated that many different variations and modifications are possible. For example, both sets of wheels may be powered by a single motor, with a suitable transfer arrangement to direct power to the appropriate set of wheels. Instead of a motor, the mechanism used to lift the wheels may use linear actuators, such as linear motors or hydraulic rams.

(46) The mechanism used to lift containers into the container-receiving space could take any suitable form. For maximum stability and load capacity, it is desirable to provide four lifting cables, with one cable disposed near each of the corners of the device, but a different arrangement, for example with fewer cables, could be used if desired. Conveniently, all of the cables are spooled and unspooled using a single motor, but more than one motor could be used if desired.

(47) It will be appreciated that that the individual lifting mechanisms may be configured so as to lift a larger container, said larger container being of equivalent footprint to an integer number of containers.

(48) Other variations and modifications not explicitly described above will also be apparent to the skilled reader.

Claims

1. A load handling device for retrieving, transporting, and delivering containers in a storage system, the storage system having a grid structure formed of two substantially perpendicular sets of rails or tracks, the grid structure forming a plurality of grid spaces and located above a plurality of stacks of one or more containers, each stack arranged to occupy a footprint of a single grid space, the load handling device configured to move laterally on said sets of rails or tracks above said stacks of containers, the load handling device comprising: a wheel assembly, the wheel assembly including a first set of wheels for engaging with a first set of rails or tracks of said sets of rails or tracks to guide movement of the load handling device in a first direction and a second set of wheels for engaging with a second set of rails or tracks of said sets of rails or tracks to guide movement of the load handling device in a second direction; a container receiving space; a plurality of vehicle modules, each of the vehicle modules having a container lifting means arranged to lift a container such that the load handling device is arranged to lift a corresponding number of containers as vehicle modules into the container receiving space; wherein the load handling device is configured for lifting a plurality of containers from a plurality of stacks in a single lifting operation, and the container receiving space is configured to hold the plurality of containers, which are lifted from the plurality of stacks, such that each container in the plurality of containers is horizontally adjacent to at least one other container in the plurality of containers, wherein the load handling device further comprises an upper part and a lower part, the lower part comprising a frame structure for supporting the first and second sets of wheels of the wheel assembly; wherein one or more elements of the frame structure are moveable to raise and lower said first set of wheels with respect to said second set of wheels to facilitate engagement of either the first set of wheels or the second set of wheels with one of the first set of rails or tracks or the second set of rails or tracks, respectively; and wherein the moveable elements of the frame structure are driven by a motor housed in the upper part of the load handling device.
2. A load handling device according to claim 1, wherein the frame structure is arranged around the container-receiving space.

3. A load handling device according to claim 1, wherein the container-receiving space is bounded on four sides by the frame structure.
 4. A load handling device according to claim 1, wherein the load handling device is configured to selectively engage and disengage the first set of wheels with the first set of rails or tracks and selectively engage and disengage the second set of wheels with the second set of rails or tracks.
 5. A load handling device according to claim 1, wherein the load handling device is configured to move in a first direction along the first set of rails or tracks and in a second direction along the second set of rails or tracks, wherein the second direction is perpendicular to the first direction.
 6. A load handling device according to claim 1, wherein the upper part houses power components, control components, drive components and/or lifting and lowering components; wherein the lower part comprises the container receiving space; and wherein the lower part is arranged directly below the upper part.
 7. A load handling device according to claim 6, wherein the first and second sets of wheels and each of support pieces associated with the first and second sets of wheels, linkages associated with the first and second sets of wheels and drive belts associated with the first and second sets of wheels are arranged around the edges of the container receiving space so that the upper part of the vehicle is solidly supported.
 8. A load handling device according to claim 1, wherein the device comprises an external housing that encloses the container receiving space.
 9. A load handling device according to claim 8, wherein the housing is shaped substantially as a cuboid.
 10. A storage system comprising: a grid structure formed of two substantially perpendicular sets of rails or tracks, the grid structure forming a plurality of grid spaces; a plurality of stacks of one or more containers located beneath said sets of rails or tracks, each of the plurality of stacks arranged such that each stack occupies a footprint of a single grid space; and the load handling device as claimed in claim 1.
 11. A storage system according to claim 10, including: a port location, the port location having container receiving means, the container receiving means including: a container receiving platform; a transferring means; and wherein the container receiving platform and the transferring means together being adapted to receive a plurality of containers delivered by the load handling device and transfer the plurality of containers to a shipping trolley.
 12. A storage system according to claim 11, wherein the transferring means comprises a push plate or a lift mechanism.
 13. A storage system according to claim 12, wherein the container receiving platform comprises at least one moveable portion such that the container receiving platform is configured to compensate for differences in the footprint pitch of the stacks and the shipping trolley.
 14. A method of transferring a plurality of containers located within a plurality of adjacent stacks in a storage system from the storage system to a shipping trolley via a load handling device according to claim 1, the method comprising: retrieving, by the load handling device, a plurality of containers from a correspondingly numbered plurality of stacks, wherein the load handling device retrieves the plurality of containers using the container lifting means of the plurality of vehicle modules in a single operation, each of the plurality of lifting means configured to lift a container of the plurality of containers into the container receiving space of the load handling device; transporting, by the load handling device, the plurality of containers from the plurality of stacks; delivering, by the load handling device, the plurality of containers to a port location; and lowering, by the load handling device, the plurality of containers via the plurality of lifting means on to a container receiving platform, wherein the container receiving platform transfers the plurality of containers to the shipping trolley.
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